

# Assignment - 02

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Prathmesh Gore

## Part A Question 01

a)  $5 \rightarrow$  scalar

b)  $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \rightarrow$  vector

c)  $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow$  matrix

d)  $(256 \times 256 \times 3) \rightarrow$  Tensor

## Question 02

a)  $2 \times 3$

b)  $4 \times 1$

~~c)  $4 \times 2$~~  c)  $3 \times 2$

## Question 03

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 7 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 2+4 & 5+1 \\ 3+6 & 7+8 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 6 \\ 9 & 15 \end{bmatrix}$$

$$A-B = \begin{bmatrix} -2 & 4 \\ -3 & -1 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 10 \\ 6 & 14 \end{bmatrix}$$

Question 05

$$\text{Transpose} \rightarrow \begin{bmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \end{bmatrix}$$

Question 04

Transpose  $\rightarrow$

$$\begin{bmatrix} 2 & 8 \\ 4 & 10 \\ 6 & 12 \end{bmatrix}$$

Part B

Question 06

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 22 \\ 43 & 50 \end{bmatrix}$$

Question 07

$$a) (2 \times 3) (3 \times 4)$$

Possible Result =  $(2 \times 4)$

$$b) (3 \times 2) (4 \times 3)$$

Not possible

$$c) (4 \times 2) (2 \times 5)$$

Possible Result =  $(4 \times 5)$

$$d) (5 \times 4) (4 \times 2) \text{ possible } (5 \times 2) = \text{Result}$$

Question 8

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 \times 5 + 1 \times 7 & 2 \times 6 + 1 \times 8 \\ 4 \times 5 + 3 \times 7 & 4 \times 6 + 3 \times 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 17 & 20 \\ 41 & 48 \end{bmatrix}$$

Question 9

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

$$BA = \begin{bmatrix} 5 \times 1 + 6 \times 3 & 5 \times 2 + 6 \times 4 \\ 7 \times 1 + 8 \times 3 & 7 \times 2 + 8 \times 4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$$

$$\text{So } AB \neq BA$$

Conclusion - Matrix Multiplication is not commutative

Part - C

$$\text{Question 10} \\ X = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{Magnitude} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Question - 11

$$y = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$

$$\|y\| = \sqrt{4 + 25 + 36} = \sqrt{65} = 8.06$$

Question 12

$$x = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$$

$$x \cdot y = 10 + 8 + 3 = 21$$

Question 13

$$x = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix}$$

$$x \cdot y = 2 - 2 = 0$$

Conclusion, dot product is zero so vectors are orthogonal

Question 14

$$x = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\|x\| = \sqrt{6^2 + 8^2} = \sqrt{100}$$

$$\|x\| = 10$$

$$\text{normalized vector} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

Question 15

$$A = (2, 3) \quad B = (8, 11)$$

$$= \sqrt{(8-2)^2 + (11-3)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

Part D

Question 16

$$y_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$\text{Since, } y_1 = 2y_2$$

They are linearly dependent.

Question 17

Need

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Equ<sup>n</sup>

$$a + 2b = 5$$

$$a + b = 7$$

$$a + 2b = 5$$

$$a + b = 7$$

$$\underline{\underline{b = -2}}$$

$$\text{Then, } a + 2(-2) = 5$$

$$\cancel{a + 2(-2)} \quad a - 4 = 5$$

$$\underline{\underline{a = 9}}$$

Hence, Vector belongs to span

Question 18

$$R^2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

Question 19

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Since, the 2<sup>nd</sup> row is twice the first

$$\text{Rank} = 1$$

Question 20

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Rows are independent, so Rank = 2

Question 21

Part E

$$a) \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Dot Product} = 0$$

Orthogonal

$$b) \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

$$\text{Dot product} = 12 - 12 = 0$$

Orthogonal

$$c) \begin{bmatrix} 2 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 2 \end{bmatrix} = 10 + 10 = 20$$

Not orthogonal

Question 22

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{Dot product} = 11$$

$$\text{Magnitude } \|x\| = \sqrt{1^2 + 2^2}$$

$$= \sqrt{5}$$

$$\|y\| = \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16} = 5$$

$$\text{Cosine} = \frac{11}{5\sqrt{5}}$$

$$\approx 0.98$$

~~Question 23~~

$\therefore$  The Vectors are highly similar & point in almost same direction

Question 23

① Orthogonal vectors  $\rightarrow$  Two Vectors whose dot product is zero

② Orthonormal Vectors  $\rightarrow$  Orthogonal Vectors whose magnitude is unit (Length = 1)

③ Basis  $\rightarrow$  A set of linearly independent vectors that can represent every Vector in a Vector Space

④ Vector Space  $\rightarrow$  A collection of vectors where vector addition & scalar multiplication are defined & satisfy the Vector Space axioms